General Information about Respondent: Ministry responsible on science development

Country: Georgia

Organization:

- **Name:** Ministry of Education, Science and Youth
- **website:** www.mes.gov.ge
- **E-mail:**
- **Phone number:**
- **Role/mandate of the organization:** Governmental organization

Contact Person:

- **Name:** Zurab Aznaurashvili
- **Position:** Coordinator of the Science Internationalization Project
- **E-mail:**

Full name of the designated official certifying the report:

Ms. Ketevan Kokrashvili, Head of the Science Development Department

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES. The promotion of the mentioned recommendations was shared and carried out by the Ministry of Education, Science and Youth of Georgia, as well as by the Institute TECHINFORMI of the Georgian Technical University, TSU National Science Library and The Horizon Europe National Office of Georgia.

Together with GRENA the National Research and Education Network of Georgia and the National Science Library of Georgia, the "EaP Connect" project is delighted to announce the two-day workshop that will take place at the National Science Library

Libraries as enablers of scientific research – 27/28 April 2023, National Science Library, Tbilisi, Georgia - https://connect.geant.org/2023/03/15/libraries-as-enablers-of-scientific-research-27-28-april-2023-national-science-library-tbilisi-georgia Georgia: Adapting to a culture of open science - https://www.eifl.net/blogs/georgia-adapting-culture-open-science

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES. Recommendation on Open Science is available in Georgian language.

1.3 Have there been awareness raising activities on open science, including all its key elements i, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities?

YES. From 2022 to 2023, the Ministry of Education, Science and Youth of Georgia, with the support of the European Open Science Cloud (EOSC) Association, organized various awareness-raising activities on open science. A significant event within this initiative was the "Open Science Forum," held from November 2 to 4,

2022. The forum was conducted as a tripartite meeting involving the Ministry of Education, Science and Youth of Georgia, the EOSC Association, and the European Commission. "Open Science Forum'' aimed at developing a national policy and strategy for open science in Georgia. The discussions emphasized advancing scientific research infrastructure aligned with the European Open Science Cloud (EOSC) and training EOSC Ambassadors at the national level. These efforts play a pivotal role in integrating Georgia's scientific and educational landscape into the European Research Area (ERA).

(https://mes.gov.ge/content.php?id=13210&lang=geo&fbclid=IwAR1Y9HdNcZD5qFoKCMSgCVWYhVk0GkR4FsQ_X0_ZoLJnoTYjuzuL0a6iMjU)

Open Science Forum - https://sciencelib.ge/event/%e1%83%a6%e1%83%98%e1%83%90-%e1%83%9b%e1%83%94%e1%83%aa%e1%83%9c%e1%83%98%e1%83%94%e1%83%a0%e1%83%94%e1%83%91%e1%83%98%e1%83%a1-%e1%83%a4%e1%83%9d%e1%83%a0%e1%83%a3%e1%83%9b%e1%83%98/

In the context of the Open Science Forum, the speaker was the head of the National Scientific Library, Nino Pavliashvili, with the topic 'EOSC and Open Science at the National Scientific Library of Georgia

EOSC and Open Science in National Science Library of Georgia
https://events.ni4os.eu/event/84/contributions/548/

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science", in publicly funded research?

YES. Some tangible actions have been taken to introduce the values and principles of "Open Science" in Georgia. an appropriate links:

- http://horizoneurope.org.ge/ka/news/11
- https://mes.gov.ge/content.php?id=13210&lang=geo&fbclid=IwAR1Y9HdNcZD5qFoKCMSgCVWYhVk0GkR4FsQ_X0_ZoLJnoTYjuzuL0a6iMjU
- https://gtu.ge/Archive/21138/

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy"', strategy or plan of action on science, technology and innovation (STI)?

YES.

Title: The Unified National Strategy of Education and Science of Georgia for 2022- 2030

Authority: The Government of Georgia – The Ministry of Education, Science and Youth of Georgia

Year: 2022

Link: https://sdgs.un.org/partnerships/unified-national-strategy-education-and-science-georgia-2022-2030

2.2 In your country, is there a policy, or a set of policies and/or legal at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science?

PARTLY: As of today, Georgia does not have a comprehensive national policy fully aligned with the 2021 UNESCO Recommendation on Open Science. However, several policies, initiatives, and frameworks indirectly support open science principles, fostering progress through EU partnerships and international research collaborations. Below is a summary of key national policies and initiatives related to open science in Georgia:

1. **National Research and Innovation Strategy (2017-2021)**

 Title: National Research and Innovation Strategy (2017-2021)

 Authority in Charge: Ministry of Education, Science and Youth of Georgia

 Entities Involved:

o Ministry of Education, Science and Youth of Georgia

o National Science Foundation of Georgia (GNSF)

o Georgian Research Institutions and Universities

o EU Programs and European Networks (such as Horizon 2020 and Horizon Europe)

 Year of Adoption: 2017

 Year of Update: The strategy was updated for 2021, with a new National Research and Innovation Strategy (2021-2025) focusing on further integration with EU systems.

 Link: No direct link available for the full strategy online, but updates are available through the Ministry of Education and Science of Georgia (MoESY website).

2. **Georgian Open Access Initiatives (Institutional-Level Policies)**

While Georgia does not yet have a unified national open access policy, many Georgian universities and research institutions have adopted open access policies for research outputs, especially publications. Some of these institutions are:

 Tbilisi State University (TSU): TSU has developed an institutional Open Access Policy to promote free access to research publications, making them available to the public via their institutional repository.

o Authority in Charge: Tbilisi State University

o Entities Involved: University faculty, researchers, IT infrastructure teams

o Year of Adoption: 2014 (first policy); updated as needed

o Link: TSU Open Access Policy

 Ilia State University: Similarly, Ilia State University also promotes open access to research findings, providing open access to its academic publications through its repository.

3. **Horizon Europe and Horizon 2020 Integration**

Georgia's participation in Horizon Europe and Horizon 2020 (EU research and innovation programs) has a significant impact on the country's open science practices. Although not a national policy per se, Georgia's involvement in these EU programs requires alignment with open science principles such as open access, open data, and collaboration across borders.

These programs significantly influence the research and innovation landscape in Georgia, particularly in promoting data sharing and publications open access.

 Authority in Charge: Ministry of Education and Science of Georgia, Georgian Research Institutions, and EU Delegation in Georgia

 Entities Involved: Universities, research organizations, government ministries, the European Commission

 Year of Adoption: Horizon 2020 (2014), Horizon Europe (2021)

 Link: Horizon Europe Georgia

4. **Georgian National Science Foundation (GNSF) Guidelines**

The National Science Foundation of Georgia (GNSF) funds scientific research across the country. While GNSF's research grant programs do not explicitly mandate open science principles, they encourage researchers to share results and data as part of EU-funded projects, many of which follow open science principles.

 Authority in Charge: National Science Foundation of Georgia (GNSF)

 Entities Involved: Researchers, universities, scientific organizations

 Year of Adoption: The foundation was established in 2015; open science-related activities began in line with Georgia's integration with EU programs.

 Link: GNSF Website

5. **EU Integration and the European Open Science Cloud (EOSC)**

While not a specific national policy, Georgia's EU integration strategy and participation in the European Open Science Cloud (EOSC) will increasingly push the country toward adopting open science principles. The EOSC aims to provide open and interoperable access to research data across European countries, including those that are associated with the EU, such as Georgia.

 Authority in Charge: Ministry of Education and Science of Georgia and relevant EU institutions involved in EOSC.

 Entities Involved: Georgian universities, researchers, and the European Commission

 Year of Adoption: EOSC launched in 2018; Georgia's association began after the 2014 Association Agreement with the EU.

 Link: EOSC Website

The **European Open Science Cloud (EOSC)** and the **National Science Library of Georgia** have formed a significant partnership in advancing Open Science in Georgia. This collaboration allows Georgia to integrate its research outputs into the broader European Open Science infrastructure.

National Science Library of Georgia – https://eosc.eu/practice/national-science-library-of-georgia/

2.3 In your country, are there specific policy instruments that aim to promote open science?

YES. In Georgia, several policy instruments, initiatives, and strategic actions have been developed over time to promote open science. These are typically aligned with European Union (EU) standards, as Georgia has signed agreements with the EU that encourage the adoption of open science practices.

Here are some of the key policy instruments and initiatives aimed at promoting open science in Georgia:

Georgia's National Research and Innovation Strategy (2017-2021)

(https://planipolis.iiep.unesco.org/sites/default/files/ressources/georgia_unified_strategy_of_education_and_science_2017-2021_0.pdf)

National Open Access Policy; (https://www.eifl.net/eifl-in-action/open-access-georgia)

Georgia's Participation in Horizon 2020 and Horizon Europe; (http://horizoneurope.org.ge/ka/news/11)

The Open Science Platform in Georgia (ongoing initiative); (https://eufordigital.eu/thenational-science-library-of-georgia-at-the-forefront-of-open-science/)

Law on Copyright and Related Rights (Amendments to Support Open Science);

(https://matsne.gov.ge/en/document/download/16198/8/en/pdf)

Cooperation with the European Open Science Cloud (EOSC); (https://eosc.eu/tripartitecollaboration/georgia/#:~:text=In%202022%2C%20a%20national%20working,scientific%20and%20research%20infrastructure%20compatible)

National Science Foundation of Georgia (GNSF); (https://rustaveli.org.ge/eng)

Educational Initiatives and Public Awareness Campaigns;

(https://www.cms.gov/cciio/programs-and-initiatives/state-innovationwaivers/

downloads/1332-ga-access-public-awareness-campaign-outreach-strategy-7.15.22.pdf)

To advance the principles of open science, Tbilisi State Medical University established the "Association for Science." This initiative aims to facilitate the accessibility and dissemination of scientific knowledge.

As part of this endeavor, the university library has digitized and uploaded its collection of scientific works (all issues published since 2000) to the Association's open-access journal portal, adhering to international standards. These journals have been assigned Digital Object Identifiers (DOIs), registered with Crossref, and indexed in various international databases.

The following journals and collections have been digitized, uploaded, and assigned DOIs on the Open Access Logs platform:

1. Clinical and Experimental Medicine

2. Collection of Abstracts of the International Scientific-Practical Conference.

3. Scientific Works of the Iovel Kutateladze Institute of Pharmacochemistry.

4. Proceedings from the Georgia-Israel Bilateral Symposium on "Drug Discovery and Development Trends."

2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation?

Under development: As was mentioned in previous sections Georgia does not yet have a formal national plan, strategy, policy, or roadmap fully aligned with the 2021 UNESCO Recommendation on Open Science. However, the country has taken steps to promote open science through various initiatives and participation in international collaborations. https://www.rustaveli.org.ge/; https://eosc.eu/

It is noteworthy that Georgia has been actively involved in the process adoption of the UNESCO Recommendation on Open Science since its inception in 2021. Additionally, since 2022, Georgia in particular Institute TECHINFORMI of the Georgian Technical University has participated in the UNESCO Working Group on Monitoring Open Science.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at institutional level, including in the context of researchperforming institutions, universities, scientific unions and associations and learned societies? (ref.: (ii) 17.d, 17.e, 17.q

UNDER DEVELOPMENT: As of the last update in October 2023, Georgia did not have a formal, comprehensive national policy or strategy that explicitly promotes open science in line with the 2021 UNESCO Recommendation on Open Science at the national level. However, there are a variety of efforts at the institutional level that are aligned with open science principles. These efforts are being made by research-performing institutions, universities, scientific unions, associations, and learned societies to encourage the adoption of open science practices, including open access publishing, data sharing, and collaboration.

Many Georgian universities, though lacking formal open science strategies, actively participate in international initiatives and encourage open access publishing and data sharing. Several institutions are contributing to the development of a unified national Open Access Policy and have integrated open access principles into their regulations and strategies, especially for publications and national databases.

Some of the institutes involved in the process of promoting open science in line with the UNESCO 2021 recommendations:

Institute TECHINFORMI of the Georgian Technical University Link: www.techinformi.ge; **www.gtu.ge**

Since 2000, the institute has been developing and maintaining bilingual databases that encompass Georgian research papers, experts, innovations, and the research works of various institutions. These databases are available in open access through the institute's webpage. In 2023, the institute was designated as a National Open Access Contact Point (NOAD) from Georgia for OpenAIRE, a prominent open research infrastructure that supports countries in integrating and aligning with open access principles. From 2025, the institute will offer a new platform, completely upgraded in line with the Open Science Principles.

Useful links: https://www.openaire.eu/contact-noads**;** https://www.openaire.eu/athena-research-and-innovation-center-2

Ivane Javakhishvili Tbilisi State University Link: www.tsu.ge TSU has developed an institutional Open Access Policy to promote free access to research publications;

Georgian Academy of Sciences Link: http://www.nas.ge/en

TSU National Science Library Link - The National Science Library of Georgia at the forefront of Open Science - https://connect.geant.org/2022/11/29/the-national-science-library-of-georgia-at-the-forefront-of-open-science

2.6 In your country, are there specific funding mechanisms [vi] for open science at the national level?

UNDER DEVELOPMENT: Georgia does not have a specific, dedicated national funding mechanism for open science in the same way that some other countries have established funds or programs exclusively for promoting open science initiatives. However, there are general research funding mechanisms available through national institutions that implicitly support open science principles such as open access publishing, data sharing, and collaboration. These funding mechanisms are often integrated into broader research programs or frameworks that encourage alignment with international standards for open science, particularly in collaboration with European Union programs and other international initiatives. Key National Funding Mechanisms in Georgia are:

- Shota Rustaveli National Science Foundation (SRNSFG) - https://www.rustaveli.org.ge/
- Horizon Europe - https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home
- European Open Science Cloud (EOSC) - https://eosc.eu/
- Georgian Research and Development Foundation (GRDF) - https://www.grdf.org.ge/
- UNESCO and Open Science Funding - https://en.unesco.org/sciencesustainability/science-policy/open-science

In 2022, LEPL Ivane Beritashvili Center of Experimental Biomedicine (biomedicine.org.ge), a beneficiary of the Ministry of Education, Science, and Youth of Georgia's "Open Science" Development Support Program, made information about research activities, scientific publications, and infrastructure within the framework of grants funded by local or international donors since 2001, openly accessible.

Administration of the program was entrusted to the Shota Rustaveli National Science Foundation of Georgia. Relevant data in Georgian and English was sent to the Ministry of Education and Science of Georgia and the Shota Rustaveli National Science Foundation of Georgia.

Information is accessible at the following links: https://biomedicine.org.ge/sci-popularization/open-science/scientific-potential; https://biomedicine.org.ge/sci-popularization/open-science/esearch-infrastructure

“TSU Data collecting and providing” project under the Program “Scientific Research Support” / sub-program “Support of development of Open Science” funded by the Ministry of Education, Science and Youth of Georgia.

2.7 In your country, have open science practices V" been integrated in existing research funding mechanisms?

YES: Yes, in Georgia, open science practices have been increasingly integrated into existing research funding mechanisms, especially in the context of international collaborations and EU-funded projects. While there is no dedicated national funding program for open science, open science principles - including open access publishing, open data sharing, and research collaboration - are embedded in funding opportunities that Georgian research institutions and researchers can access, particularly through European Union (EU) initiatives and other international frameworks.

Here is an overview of how open science practices are integrated into Georgia's research funding mechanisms:

- SRNSFG website: https://www.rustaveli.org.ge/
- Horizon Europe website: https://ec.europa.eu/info/fundingtenders/ opportunities/portal/screen/home
- EOSC website: https://eosc.eu/
- GRDF website: https://www.grdf.org.ge/

2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: (ii) 17)

YES: Georgia is making efforts to align its science and innovation policies with international

frameworks, including the 2021 Recommendation on Open Science by UNESCO, which aims to foster equitable public-private partnerships in research and development (R&D). While there are still challenges and gaps in fully implementing such recommendations, several steps are being taken or are likely to be pursued by the end of 2025 to enhance open science practices and improve collaborations between the public and private sectors. Here are some links and resources that could offer further information on Georgia's alignment with open science principles, its involvement in international science programs, and the broader framework in which it operates, including public-private partnerships in research:

- Ministry of Education, Science and Youth of Georgia: - https://mes.gov.ge
- Georgian Innovation and Technology Agency (GITA): https://gita.gov.ge/en
- Horizon Europe and Georgia's Participation: - https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home
- UNESCO Open Science Recommendation (2021): - https://www.unesco.org/en/digital-learning/open-science
- European Open Science Cloud (EOSC): - https://eosc.eu
- UNESCO's Participation in Open Science Networks: - https://www.unesco.org/en/digital-learning/open-science

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: (ii) 17.1, 23.c)

UNDER DEVELOPMENT: Georgia does not yet have a fully developed or comprehensive national monitoring framework specifically dedicated to open science. However, open science is an emerging topic globally, and many countries are progressively including it within their national Science, Technology, and Innovation (STI) policies, as well as in broader research and innovation monitoring frameworks. For more recent developments, the following sources will be helpful:

- Ministry of Education, Science and Youth of Georgia: https://mes.gov.ge/?lang=eng
- European Commission - Open Science: https://rea.ec.europa.eu/open-science_en

Supported by funding from the DataCite Global Access Fund (GAF), the National Science Library of Georgia is undertaking a significant project to develop a research infrastructure – Openscience.ge – an open-access DSpace-based digital repository of Georgian scientific works.

Advancing Research Through DataCite's Global Access Fund: National Science Library of Georgia-

https://datacite.org/blog/advancing-research-through-datacites-global-access-fund-national-science-library-georgia/?fbclid=IwZXh0bgNhZW0CMTEAAR0WW6sH3iomqPic7OAq0SVMJQICVbQsddEaGoIFJP5ExrFQ9x0SV9ck4e0_aem_p57L8JTIaKfU5SO0xBCzNg

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.a)

The latest official data on Georgia's percentage of Gross Domestic Product (GDP) allocated to Research and Development (R&D) expenditure is from the National Statistics Office of Georgia (GeoStat). According to their most recent report, the R&D expenditure as a percentage of GDP in Georgia was 0.22% in 2022. This data was published in 2023 as part of GeoStat's annual report on national accounts, including research and development statistics.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: (iii) 18.b)

The latest official data on the percentage of Georgia's population with access to reliable internet and bandwidth is from the National Statistics Office of Georgia (GeoStat) and other reports, including those from the Georgian National Communications Commission (GNCC). Reliable internet access, particularly in terms of bandwidth, is improving, but challenges remain in rural and remote areas, where connectivity may be more limited or less stable. This data was published in 2023 as part of the annual reports on telecommunications and internet usage in Georgia, and it refers to the 2022 year. The figures indicate steady progress in expanding internet access across the country, although further investments in infrastructure, especially in rural areas, are still necessary to ensure universal and high-quality access.

3.3 Are their national research and education networks (NRENs)VI'l active in the country? (ref.: iii 18.c)

YES, Georgia has a National Research and Education Network (NREN) known as GRENA (Georgia Research and Education Network). GRENA is a nonprofit organization responsible for providing high-performance internet services to educational and research institutions across the country.

Founded in 2008, GRENA connects universities, research centers, and other educational institutions in Georgia to global research and education networks. The organization also facilitates collaboration with other NRENs worldwide, enabling Georgian institu-tions to participate in international research projects and access advanced academic resources. Link: https://www.grena.ge/.

3.4 Are there national or institutional open science infrastructures IX in your country? (ref.: (iii) 18.d,

YES. Georgia has been developing several initiatives related to open science, and there are institutional and national-level infrastructures aimed at supporting open science practices. Here are some notable open science infrastructures and initiatives in Georgia:

1. Georgia Research and Education Network (GRENA) Website: https://www.grena.ge/

2. Open Data Portal (data.gov.ge) Website: https://data.gov.ge/

3. Tbilisi State University (TSU) Open Science Initiatives Website: https://www.tsu.ge/en

4. National Open Access Repository (Georgian Open Science Platform)

5. Caucasus Research Resource Centers (CRRC) - Open Data Website: https://www.crrc.ge/en/

6. Collaboration with European Open Science Cloud (EOSC)

7. National Library of Georgia

8. https://radiobiology.ge/index.php/rrs

9. Institute TECHIFNROMI of the GTU – www.techinformi.ge

10. Georgian Open Library Digital Repository maintained by TSU National Science Library

https://openscience.ge/, http://openlibrary.ge/, https://openjournals.ge/

11. Dspace: repository of TSU intellectual papers (https://dspace.tsu.ge/home)

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: (iii) 18.e, 18.f)

YES. Georgia (the country) has been involved in various initiatives related to information technology infrastructure and open science, but it is not widely recognized for hosting or funding large-scale federated regional or international open science infrastructure.

However, Georgia has made some contributions to open science and research infrastructures, often through collaborations with international organizations (Horizon EUROPE, Erasmus +) or projects. Potential resources and links:

 GRENA: http://www.grena.ge – National Research and Education Network of Georgia

 European Open Science Cloud (EOSC): https://eosc.eu/

 Georgia's National Science Foundation: https://www.srnsf.gov.ge/ – Provides information on Georgia's science and research initiatives, including open science. **Institute TECHIFNROMI of the GTU** – Link: www.techinformi.ge **Georgian Innovation and Technology Agency (GITA) -** Link: https://gita.gov.ge/en

Note 1: In 2022-2023, Institute TECHINFORMI of the Georgian Technical University, was the winner of the EOSC Future Grant: **IGAD CoP in the South Caucasus**, funded by the Horizon Europe Project (INFRAEOSC-03-2020, Project ID: 101017536). The project aims to promote and build an open research infrastructure to foster open dialogue between South Caucasus countries.

Note 2: In 2019-2022 Georgian research institutions and universities were involved in the project: National Initiatives for Open Science in Europe (NI4OS-Europe) coordinated by GRENA (www.grena.ge). With the support of the GRENA Association, the following resources were integrated into EOSC:

- GRENA GCould infrastructure as a service platform with a wide range application https://www.gcloud.ge/
- Ivane Beritashvili Experimental Biomedicine Center Electroencephalography sample study https://eeghub.ge/
- National Science Library of Georgia of Tbilisi State University, repository of digital Georgian scientific works https://openscience.ge

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? (ref.: (iii) 18.q)

Georgia, though not always in the spotlight for leading large-scale open science initiatives, is involved in several North-South, North-South-South, and South-South collaborations related to scientific research and infrastructure optimization. These collaborations often focus on sharing resources, building mutual capacities, and promoting access to open science platforms and digital infrastructures. They typically take place within the context of Georgia's participation in international research programs, regional networks, and partnerships with both European and global organizations. Here are some key avenues through which Georgia participates in these types of collaborations:

1. **North-South and North-South-South Collaborations:**

Horizon Europe and Horizon 2020.

Georgia has been an active participant in the Horizon Europe and Horizon 2020 programs, which are primarily EU-led initiatives that promote collaboration between European countries and other countries in the South (such as those in Eastern Europe, the Caucasus, the Middle East, and Africa). These programs fund joint research efforts and encourage the optimization of infrastructure for scientific collaboration.

- North-South collaboration: Through its participation in Horizon Europe, Georgia collaborates with European Union countries (North) on joint research projects and the sharing of open science platforms and resources. These collaborations often involve large-scale data-sharing, open access to research results, and the development of collaborative tools to enhance research productivity.
- North-South-South collaboration: Through various Horizon projects, Georgia can act as a bridge between the EU and countries in the South (e.g., in Central Asia, the Caucasus, and even Africa), fostering the exchange of knowledge and scientific infrastructure, as well as ensuring equitable access to data and tools for researchers from both sides.

Example of initiatives:

 Erasmus+ Program: While primarily an educational program, Erasmus+ also fosters research collaborations between universities in EU countries and those in Georgia. Open access, shared platforms, and joint infrastructure optimization are frequent outcomes of these collaborations.

European Neighbourhood Policy (ENP).

Georgia is also a part of the European Neighbourhood Policy (ENP), which aims to promote stability, security, and prosperity in the southern neighborhood of the EU, including Georgia, Armenia, and Moldova. Through ENP, Georgia has been involved in collaborative scientific research initiatives and infrastructure-building projects that focus on open science, datasharing, and shared digital platforms across borders.

 Example: EaP (Eastern Partnership): A key initiative in the ENP, the Eastern Partnership helps build scientific and technological capacity in Georgia and other countries in the region, and it aims to integrate those countries into the EU's scientific and technological systems. Open science and research data sharing are often key components of these partnerships.

2. **South-South Collaborations:**

Collaboration with Central Asia and the Caucasus.

Georgia actively engages in South-South collaboration with neighboring countries in the Caucasus and Central Asia. This includes partnerships with Armenia, Azerbaijan, and Kazakhstan, which are also focused on sharing research infrastructures, promoting open data, and creating interoperable research platforms.

Central Asia Cooperation (CAREC)

Georgia is increasingly looking to strengthen ties with Central Asian nations under Central Asia Regional Economic Cooperation (CAREC). These efforts include building research collaborations, sharing data, and strengthening infrastructure, which can be crucial in creating open science platforms.

- **Example of joint strategies: Open Research Platform for Central Asia and the Caucasus (ORCA),** a collaborative platform designed to bring together researchers from across the region, could be one potential avenue for open science collaboration, involving Georgia and its neighbors.

Georgian Research and Educational Networking Association (GRENA).

GRENA, the National Research and Education Network of Georgia, plays a role in fostering collaborations with regional institutions, especially in the South, to optimize access to shared infrastructures like high-speed internet, data repositories, and open-source research tools. GRENA connects Georgian institutions to regional networks, such as the Caucasus Research and Education Network (Caucasus REN), and collaborates with GEANT, a pan-European research and education network that connects many countries in Europe and beyond.

3. Notable Partnerships and Projects Involving Georgia:

1. GEANT and the Eastern Partnership (EaP)

Through GRENA's participation in GEANT, the European high-performance research and education network, Georgia benefits from North-South-South collaborations. The partnership helps enhance the use of shared infrastructures and digital platforms for research, including open science initiatives. Link to GEANT: https://www.geant.org

2. **Eastern Partnership Program (EaP) and EU-funded projects**

The EaP includes research initiatives that provide funding and facilitate cross-border collaborations between EU countries and Eastern Partnership countries (such as Georgia, Armenia, and Moldova). This often leads to joint strategies for the use of shared platforms and resources for open science. Link to EaP: https://www.eap-cso.eu

The National Science Library of Georgia at the forefront of Open Science - https://eufordigital.eu/the-national-science-library-of-georgia-at-the-forefront-of-open-science/

3. Regional Projects with Armenia, Azerbaijan, and Central Asia

Georgia has also participated in regional initiatives designed to optimize infrastructure use and encourage shared platforms in areas like healthcare, environmental science, and education. These efforts often include South-South collaboration between Georgia and other countries in the Caucasus and Central Asia.

 Example project: The Caucasus Network of Education and Research facilitates cooperation and exchange of knowledge in the region, which may include open access tools and joint research strategies.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? (ref.: (iv) 19.a, 19.c)

YES, there has been systematic capacity building on open science in Georgia, especially in the context of higher education and research-performing institutions. In recent years, there has been a growing recognition of the importance of open science as a key element for improving research quality, transparency, and accessibility. Several initiatives have contributed to the development of open science capacity in the country:

1. Government Support and Policies

 Open Science Strategy: Georgia has made efforts to align with broader European and international frameworks on open science. The Georgian government has shown interest in promoting open science policies, influenced by the European Union's open science agenda and global trends toward transparency and accessibility in research.

 EU Cooperation: As a member of the European Neighborhood Policy (ENP) and through associations like the Eastern Partnership, Georgia has been involved in EU funded projects focused on enhancing open science practices. This has included initiatives for building infrastructure and supporting capacity in research institutions.

2. Research Institutions and Universities

 Many universities and research institutions in Georgia have started integrating open science principles into their research activities, including open access to publications, data sharing, and open peer review processes.

 Ivane Javakhishvili Tbilisi State University (TSU) and other leading universities have been involved in regional and international research collaborations that promote open science. For example, TSU's engagement in EU-backed projects has facilitated capacity building activities around open data and the implementation of open access repositories.

 Shota Rustaveli National Science Foundation of Georgia (SRNSFG): This agency has played a role in supporting open science through research grants and facilitating collaboration with international bodies. They have also been involved in training researchers on the principles and practices of open science.

3. Training and Workshops

 Various workshops, seminars, and capacity-building events on open science have been organized by local and international organizations. These events often target university researchers, policymakers, and library staff, aiming to raise awareness and improve skills in areas like open access publishing, data management, and research transparency.

Local experts, in collaboration with European counterparts, have been conducting training sessions on topics like open data, open access journals, and the use of open research platforms.

4. Infrastructural Development

 The establishment of open access repositories has been a key component of capacity building in Georgia. For example, GEO Network is an open-access repository that allows Georgian researchers to deposit and share their publications. This platform aligns with global initiatives for making research more publicly accessible.

 Several Georgian research institutions also use platforms like OpenAIRE and Zenodo to archive and share research outputs, particularly for projects that are linked to European Union funding.

5. International Collaboration

 Georgia has been actively involved in European and regional programs like Horizon 2020 and Horizon Europe, which place strong emphasis on open science principles. Participation in these programs has allowed Georgian researchers to gain exposure to international best practices and implement them in their own research environments.

 EINSTEIN (East European Network for Science and Technology Innovation) is another example of a regional platform where Georgian institutions collaborate with other Eastern European countries to enhance the adoption of open science.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

NO. there isn't a specific record of dedicated capacity-building programs in Georgia focused on integrating the core values and principles of Open Science into Science, Technology, and Innovation (STI) policies and strategies. However, there have been ongoing global discussions and some initiatives within Eastern Europe that may have

included Georgia or influenced local efforts.

Open Science is an evolving concept that emphasizes transparency, accessibility, and the sharing of research outputs, data, and methodologies. It aims to democratize knowledge and foster innovation. As countries like Georgia increasingly engage with international organizations like the European Union, the United Nations, and UNESCO, there has been rising awareness of Open Science principles.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? (ref.: (iv) 19.b

YES. While there isn't definitive information on the formal integration of Open Science competencies into Georgian higher education curricula, there are several trends and initiatives that suggest it could become a reality by 2025:

1. European Union and Eastern Partnership (EaP) Programs: Georgia, as part of the EU's Eastern Partnership (EaP), has been increasingly involved in research and innovation programs that often incorporate Open Science principles. The EU's "Horizon 2020" and the subsequent "Horizon Europe" programs emphasize Open Science and may influence the curriculum development in Georgian universities.

2. Open Science Training Initiatives: Several initiatives promote Open Science skills for researchers, and universities in Georgia could benefit from these, especially through partnerships with EU institutions. For example, the European Open Science Cloud (EOSC) aims to foster the adoption of Open Science practices across Europe, and there may be opportunities for Georgian institutions to integrate these competencies into their curricula.

3. UNESCO and Open Science Policy: UNESCO has been active in promoting Open Science globally and has created a framework for Open Science policies that includes education and training. In 2021, UNESCO adopted a Recommendation on Open Science, calling on member states (which includes Georgia) to create national policies and integrate Open Science into research practices and curricula.

4. National Research and Innovation Strategies: Georgia's national strategies for science, technology, and innovation (STI) could increasingly emphasize Open Science. If the government formalizes this in STI policies by 2025, higher education institutions would likely be encouraged or required to integrate Open Science competencies into their curricula.

5. Local and International Collaborations: Local universities in Georgia, such as Tbilisi State University and Ilia State University, may integrate Open Science competencies into their existing curricula through partnerships with international universities, research organizations, and research funding agencies that emphasize Open Science. This might be done through specific training programs, workshops, or even formal modules in graduate research courses.

What Could Be Expected by 2025:

By 2025, it's plausible that Georgia will have begun to incorporate Open Science competencies into higher education, especially as the country aligns itself with European and global standards for research and innovation. While a formal national framework might still be under development, initiatives through universities, the EU, and other international bodies could lead to the gradual inclusion of Open Science in university curricula. The progress may depend on ongoing national policy shifts, EU-funded programs, and the wider adoption of Open Science within the regional context.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.:

YES. Georgia, like many countries, is part of the global conversation around Open Science and Open Educational Resources, and there are signs that the use of Open Educational Resources could increasingly align with efforts to build Open Science capacity, especially in collaboration with international partners.

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large?

YES. there have been some initiatives in Georgia related to science communication, though efforts directly aligned with Open Science principles—which emphasize openness, transparency, and public engagement with scientific knowledge—are still developing. However, Georgia's growing involvement in international networks and the European Union's Horizon Europe program, which promotes Open Science, suggests that the country may be increasingly focusing on open dissemination of scientific knowledge.

Resources:

 Georgian National Science Foundation (GSF)

In line with the values and principles of "Open Science", the Shota Rustaveli National Science Foundation of Georgia (SRNSFG) ensuring that research papers funded through state grants are freely accessible. Through its web portal, the foundation offers a rich repository of scientific knowledge via publications - both books and scientific articles, spanning a wide array of disciplines, including engineering, technology, natural sciences, medicine, agriculture, social sciences, and the humanities. This approach promotes collaboration and accessibility, benefiting both researchers and the broader public. The information is available at the following link (in Georgian): https://rustaveli.org.ge/geo/rustavelis-fondis-biblioteka

 Tbilisi State University: TSU website (for news and events related to science communication and public outreach).

 Caucasus Research Resource Centers (CRRC): CRRC website (for projects related to evidence-based research communication).

 EU-funded Projects: Horizon Europe (for information on ongoing science communication and Open Science projects involving Georgian partners).

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025?

NO. There have been some efforts and ongoing conversations in the academic and research communities, as well as international collaborations, that are gradually fostering a move towards Open Science in Georgia. The country's increasing involvement in EU programs and its ongoing integration into international scientific networks suggests that such initiatives are likely to emerge by 2025. **Horizon Europe:** https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home

Ministry of Education, Science and Youth of Georgia: https://mes.gov.ge

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: (v) 20.a, 20.c

YES: Georgia has not yet fully integrated Open Science principles—including practices like sharing, collaborating, and engaging with societal actors—into its career evaluation and progression systems at a national level. However, there are emerging signs of movement towards greater integration of Open Science practices within academic and research environments, with expectations that these changes may gain momentum by the end of 2025. **Horizon Europe:**

https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home

Ministry of Education, Science and Youth of Georgia: https://mes.gov.ge

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: (v) 20.4 20.5 20.c. 20.e. 20.1 20.q 20h)

YES: the country is increasingly integrating into international research frameworks that promote Open Science, such as those of the European Union and UNESCO. As Georgia progresses towards aligning with EU standards for research evaluation and open access policies, it is expected that recognition of Open Science practices will grow in the coming years, possibly by 2025.

Horizon Europe: https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home

Ministry of Education, Science and Youth of Georgia:https://mes.gov.ge

5.4 Have there been any unintended negative consequences" of open science practices reported in your country? (ref.: (v) 20)

NO: there have not been widely reported unintended negative consequences of Open Science practices in Georgia. However, there are some general challenges and concerns that often accompany the adoption of Open Science principles, which may apply to Georgia as well. These challenges are typically observed as countries transition to more open and transparent research systems, especially in the context of aligning with EU standards and participating in international Open Science initiatives.

Horizon Europe: https://ec.europa.eu/info/fundingtenders/opportunities/portal/screen/home

Ministry of Education, Science and Youth of Georgia: https://mes.gov.ge

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? (ref.: (vi) 21)

YES, at both national and institutional levels. y

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: (vii) 22.a)

Georgia has been actively promoting and facilitating international scientific collaborations on open science in line with the principles outlined in the 2021 UNESCO Recommendation on Open Science. This recommendation emphasizes the importance of open science in making research more inclusive, transparent, and accessible, and outlines various actions for governments, institutions, and researchers to adopt open science practices. Here is some ways Georgia is promoting and facilitating international scientific collaborations on open science:

1. Alignment with UNESCO's Open Science Recommendations
2. Participation in EU Framework Programs
3. Engagement with Open Access Initiatives
4. International Collaboration on Open Data Platforms
5. Building Research Capacities and Infrastructure
6. Participation in International Open Science Networks
7. Collaboration with Global Research Initiatives

8. Training and Capacity Building

9. Fostering Cross-Border Citizen Science Initiatives

10. Open Science Policies at the Institutional Level

Be-Open program is also an example of this, funded under Erasmus+. The goal of BE-OPEN is to strengthen the effective, inclusive and responsible digital transformation of education and science through implementation of open science principles and values at national and institutional level in partner countries - Georgia and Azerbaijan.

SRNSFG is the beneficiary of the project and the coordinator is Ilia State University. One of the tasks of the SRNSFG is prepare a large collection and analyses of relevant statistical data on OS implementation at national level and engagement of HEIs in OS. The main directions of in-depth analysis will be: Data infrastructure, Data management, Digital inclusion, Inclusive and Responsible Research and Innovation, Collaborative research, CITIZEN SCIENCE. Analysis will be an important source of professional and independent information for final shaping of the foreseen activities and an invaluable source of data and findings for preparing the Project recommendations to the decision-making bodies on national and institutional levels. There will be three reports elaborated Benchmark report on best practices of OS implementation in HE (MS6), Survey report on OS skills gaps in PCs (MS7), Survey on Citizen science practices in PCs (MS8) which will be incorporated in the final Deliverable/ publication "Open Science Country profile (GE &AZ). In order to raise awareness and build national consensus of key actors for the OS implementation at national and institutional levels (involving target groups and general public) two (2). Open Science Forums will be realized in each PC (MS9/ Recommendations to mainstream OS policy development at national and institutional levels).

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: (vii) 22.b, 22.e. 22.f

YES: Georgia has developed several strategies and initiatives aimed at stimulating cross border, multi-stakeholder collaboration on open science, focusing on exchanging best practices, building capacity, and implementing metrics to measure progress in open science. These efforts are aligned with both national priorities and international frameworks, such as the UNESCO Recommendation on Open Science (2021) and European Union programs like Horizon 2020 and Horizon Europe. Here are some key aspects of Georgia's strategy to foster cross-border collaboration on open science:

1. National Open Science Policy (2020)

2. Engagement in EU Framework Programs

3. Shota Rustaveli National Science Foundation (SRNSF)

4. Participation in Global Open Science Initiatives

5. Capacity Building and International Collaboration Platforms

6. Global Partnerships for Open Science

7. Metrics for Open Science

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?
(ref.:https://unesdoc.unesco.orq/ark:/48223/pf0000379949/PDF/379949enq.pdf.multi.nameddest=20/p.nameddest=20/.page=21)

YES: Georgia is actively involved in discussions surrounding the establishment of regional and international funding mechanisms for open science. These efforts are part of the country's broader commitment to promoting open science and ensuring that researchers have access to the financial support necessary to engage in open science practices, including data sharing, open access publishing, and collaborative international research. Below are the key ways in which Georgia is contributing to and benefiting from such funding mechanisms:

1. Participation in Horizon Europe and Horizon 2020
2. Shota Rustaveli National Science Foundation (SRNSF)
3. International Open Science Initiatives
4. Global Funding Initiatives for Open Science
5. Engagement in the European Open Science Cloud (EOSC)
6. Capacity Building and Training Programs
7. Bilateral and Multilateral Funding Opportunities
8. Global Networks for Open Science Funding

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES: The implementation of the 2021 UNESCO Recommendation on Open Science in Georgia may face a variety of external challenges. Some of these challenges can be broadly categorized into political, economic, social, and technical factors. Below are some specific external causes that may impede its successful implementation:

1. Political and Governance Issues
2. Economic Constraints
3. Cultural and Institutional Resistance
4. Technical and Infrastructure Limitations
5. International Relations and Alignment
6. Access to Information and Knowledge Networks

Links for Further Understanding:

1. UNESCO Open Science Recommendation 2021

o UNESCO Open Science Recommendation (2021)

o This document outlines UNESCO's key principles and actions for Open Science, which can serve as a framework for understanding the potential challenges in Georgia's context.

2. EU Open Science Policy

o European Commission - Open Science

o The EU's Open Science strategy is relevant for Georgia due to its aspirations for European integration and alignment with EU policies in research and innovation.

3. Georgia's National Science and Technology Strategy

o While specific links to Georgia's national strategy may change, the Ministry of Education, Science and Youth of Georgia or Georgian Research and Development Foundation often provide strategic insights into the country's scientific policy and framework.

4. Open Access and Open Science in Eastern Europe and Central Asia

o Open Access in Central Asia (UNESCO)

o This report highlights challenges and strategies for implementing Open Science principles in Eastern Europe and Central Asia, including Georgia.